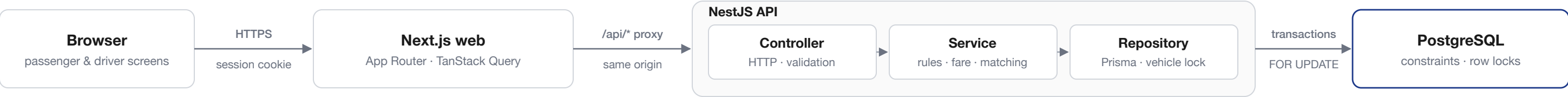
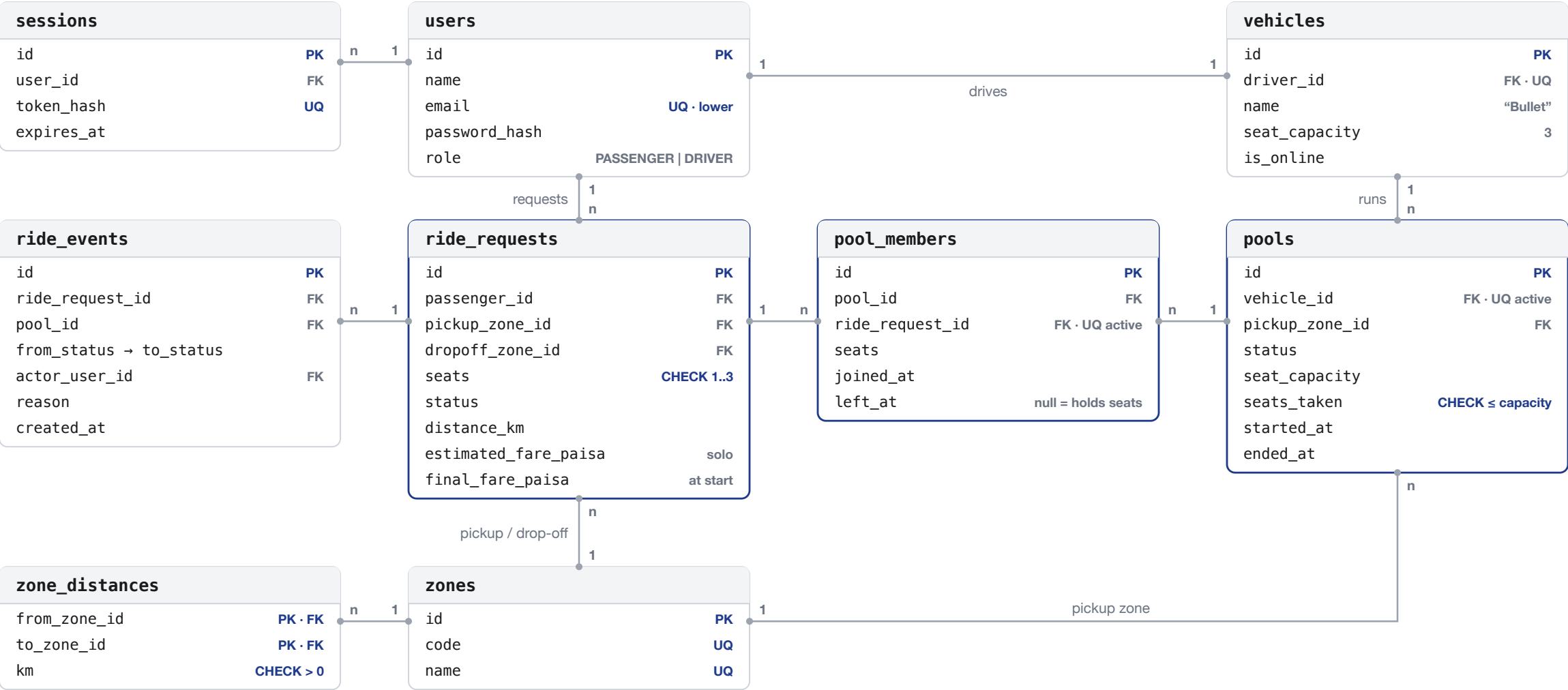


1 REQUEST PATH



Every API route passes guards: session → role → ownership. A lower layer never imports an upper one.

2 DATA MODEL



**Integrity guarantees**

- seats\_taken ≤ seat\_capacity (CHECK)
- 1 active pool per vehicle
- 1 active pool per ride request
- 1 active ride per passenger
- seat change = lock vehicle → check → write
- database rules hold even if code is wrong

**Fare**

(Tk 30 + km × Tk 15) × seats

–20% if 2+ passengers at start

estimate = solo fare, never exceeded

locked at start · integer paisa

Nusrat Tk 75 → 60 · Rafiq Tk 90 → 72

**Legend**

PK primary · FK foreign · UQ unique

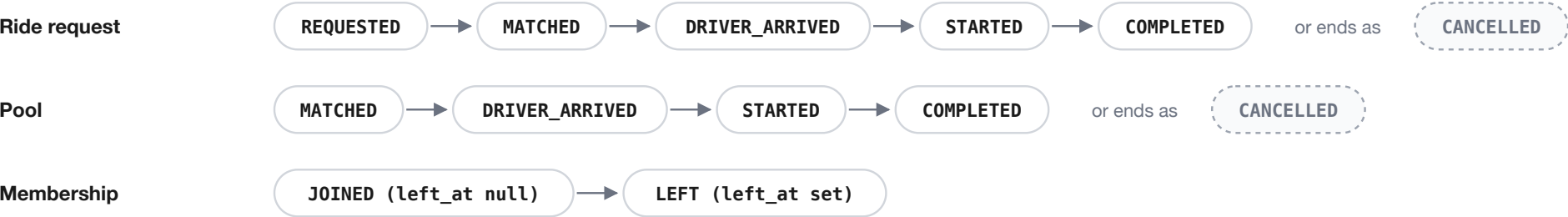
1 — n one-to-many

blue border = core of pooling

Cast: Jashim · Bullet (3 seats)

Nusrat · Rafiq · Shirin

3 LIFECYCLES



Joins (M1–M4): same pickup zone, pool still MATCHED, seats free, every detour ≤ 2 km. Any transition not shown is rejected (409); every change writes a ride\_event.